BRAVE SPARKIDS ACADEMY

Development Team Requirements

CMS, Candidate Registration & Hall Ticket Management

INTERNAL WORKING DOCUMENT

Prepared for requirements discussion and project planning

1. Project Overview

This document defines the initial functional and technical requirements for extending the existing Brave SparKids Academy website (www.bravesparkids.com) with a Content Management System (CMS), candidate/application management, and hall-ticket generation and delivery. Requirements must be validated with the client before implementation.

2. Objectives

- Allow authorised staff to update website content without developer intervention.
- Collect and manage candidate registrations for the relevant exams or assessments.
- Provide an admin workflow to review applications and approve eligible candidates.
- Generate consistent, downloadable PDF hall tickets from verified candidate and exam data.
- Give candidates a clear way to access their hall tickets and receive status notifications.
- Maintain appropriate security, privacy, auditability, and operational reporting.

3. Scope and Modules

Module	Core capabilities	Out of scope until confirmed
CMS	Manage pages, courses, FAQs, announcements, contact details, banners and SEO fields; draft/publish workflow; media uploads.	Full e-commerce, multi-language publishing, complex page builder.
Candidate Management	Registration/application form, candidate profile, application status, search/filter, verification, export.	Online payment, document verification integrations, bulk imports unless approved.
Hall Ticket Management	Exam/session setup, eligibility/approval checks, PDF generation, secure access, download status, notifications.	Automated seating allocation, QR-based attendance, SMS/WhatsApp provider fees unless confirmed.
Admin & Access	Role-based access, dashboard, audit log, settings and reports.	Complex organisation/franchise tenancy unless confirmed.

4. User Roles

Role	Permissions
Super Admin	Manage users and roles, CMS, exams, candidates, hall-ticket rules, settings and reports.
Content Editor	Create and edit website content; publish only if publishing permission is granted.
Exam/Admin Operator	Create exam sessions, review candidate applications, approve/reject, generate/reissue hall tickets.
Candidate/Parent	Submit an application, view status and access the hall ticket using the approved secure method.
Read-only/Management	View dashboards and reports without changing records.

5. Candidate Data Dictionary — Proposed

Collect only fields required by the client's process. Mark optional fields clearly and confirm the final list before form design.

Category	Proposed fields	Notes
Candidate identity	Candidate full name; date of birth or age where required; gender only if operationally necessary; candidate photo only if required.	Confirm which fields are mandatory and whether spelling rules apply.
Contact	Mobile number; email address where available; address only if required.	Verify contact details if they will be used for secure access or notifications.
Parent/guardian	Parent/guardian name; mobile number; relationship.	Required for minors if the client's process needs guardian contact.
Academic/course	School/institution; class/grade; course/program; branch/franchise if applicable.	Use controlled dropdowns where possible.
Application/exam	Application ID; exam name; exam date/session; centre; registration date; status.	Exam and centre data should be selected from admin-managed records.
Payment	Payment status and transaction reference only if fees are part of this workflow.	Do not store card details or unnecessary payment credentials.
Documents	Uploaded proof/photo/certificate only when explicitly required.	Set file type/size limits and access restrictions.
System metadata	Created/updated timestamps; status history; admin actions; hall-ticket generation/download timestamps.	Use for audit and support.

6. Candidate Statuses

- Draft (if candidates can save and return)
- Submitted
- Under Review
- Additional Information Required
- Approved
- Rejected (with internal reason and an appropriate candidate-facing message)
- Hall Ticket Generated
- Hall Ticket Reissued / Cancelled (where applicable)

7. End-to-End Workflow

1. Admin creates or updates exam/session details, eligibility rules, exam date/time, centre and hall-ticket template fields.
2. Candidate opens the registration form and submits the required details.
3. System validates mandatory fields, creates a unique application ID and records the submission timestamp.
4. Admin reviews the application and marks it approved, rejected, or requiring additional information.
5. Only eligible and approved candidates can receive a hall ticket, subject to client-confirmed rules such as payment completion if applicable.
6. System generates a PDF using a controlled template and stores the file securely or regenerates it from verified data.
7. Candidate is notified through the confirmed channel and accesses the hall ticket through an authenticated or otherwise secure method.
8. Admin can reissue or cancel a hall ticket with a recorded reason; system retains an audit trail.

8. Hall Ticket Requirements

- Unique hall-ticket number linked to the application and exam session.
- Academy branding and contact details, candidate name, application ID, photo only if required, exam name, date, time, venue/centre and reporting time.
- Instructions, permitted items, identification requirements and signature/stamp fields if supplied by the client.
- Consistent PDF layout that works on mobile and prints on A4 paper.
- Secure download route; do not expose predictable public file URLs or allow one candidate to access another candidate's ticket.
- Clear handling for missing data, duplicate generation, reissue, cancellation and exam-date changes.
- Optional QR code or barcode only if a real validation/check-in workflow is approved.

9. Proposed UX Screens

Area	Screens / key elements
Public website	Home, courses/programs, about, announcements, contact/inquiry forms, registration entry point.
Candidate journey	Registration form, confirmation page, application status lookup/login, hall-ticket page, PDF download.
Admin CMS	Dashboard, page/content editor, media manager, course/program manager, announcements, preview/publish.
Admin candidate operations	Candidate list, filters/search, candidate detail, review actions, status history, export.
Exam & hall tickets	Exam/session setup, centre configuration, approved-candidate list, generation/reissue controls, delivery status.
Administration	User/role management, notification settings, audit log and reports.

10. Security, Privacy & Reliability

- Enforce server-side validation and role-based access checks on every protected action.
- Use secure authentication, strong password handling and session/token expiry appropriate to the chosen stack.
- Limit access to children's and guardian data; do not collect sensitive or unnecessary information.
- Restrict upload types and sizes; validate files server-side and store them in non-public storage where possible.
- Protect against ID enumeration and insecure direct-object-reference access to applications and hall tickets.
- Record administrative approvals, rejections, edits, ticket reissues and cancellations in an audit log.
- Use HTTPS, backups, error logging and a documented recovery process.
- Define retention and deletion rules with the client before launch.

11. Suggested Logical Data Entities

- users, roles, permissions
- cms_pages, cms_sections/content_blocks, media_assets, announcements, courses
- candidates, guardians (if needed), applications
- exams, exam_sessions, exam_centres, eligibility_rules
- hall_tickets, hall_ticket_events or ticket_versions
- notifications, notification_attempts

- audit_logs

Entity names are conceptual. Final schema depends on the selected framework, database and confirmed requirements.

12. API / Service Boundaries

Service area	Example operations
CMS	List published pages; create/update page; publish/unpublish; upload media.
Registration	Create application; validate fields; retrieve candidate's own application; submit updates if permitted.
Review	Search/filter applications; retrieve details; approve/reject/request changes; view history.
Exam setup	Create/update exam session and centre; manage eligibility and ticket configuration.
Hall ticket	Check eligibility; generate PDF; retrieve candidate's own ticket; reissue/cancel ticket.
Notifications	Send registration confirmation, status update and hall-ticket-ready message; track delivery outcome.

13. Acceptance Criteria / QA Checklist

- Admin can create, preview, publish and update approved CMS content without code changes.
- Registration form blocks invalid submissions and produces a unique application ID.
- Admin can search and filter applications and review status history.
- Only authorised users can approve candidates or modify exam settings.
- Unapproved or ineligible candidates cannot access a valid hall ticket.
- Generated PDFs show correct candidate, exam, time and centre information.
- A candidate cannot access another candidate's application or hall ticket by changing an ID or URL.
- Reissue/cancellation actions are logged and the current ticket status is clear.
- The core flows are tested on common mobile and desktop screen sizes.
- Backup, error handling and basic operational monitoring are configured before production launch.

14. Decisions Required from Client Before Development

- Which exam(s), assessment types and candidate groups are in scope?
- Who can register: candidate, parent/guardian, school, franchise, or admin?
- Exact mandatory and optional fields, and required documents.
- Whether payment is involved and whether payment status controls eligibility.
- Who verifies applications and what approval rules apply?
- Hall-ticket fields, branding, instructions, QR/barcode requirement and print format.
- Candidate access method: login/OTP, application ID plus verification, or another approved approach.
- Notification channels: email, SMS, WhatsApp or manual sharing; provider and cost responsibility.
- Exam-centre assignment rules and whether seating/room allocation is required.
- Expected candidate volume, bulk operations, exports and reporting requirements.
- Preferred technology, hosting, backup, support and maintenance expectations.
- Data retention, privacy notices and consent requirements.

15. Delivery Plan — Proposed

9. Requirements sign-off and final field/status dictionary.
10. UX wireframes and hall-ticket template approval.
11. Database/API design and role/permission definition.
12. CMS and admin foundation.
13. Candidate registration and review workflow.
14. Exam setup, PDF hall-ticket generation and secure access.
15. Notifications, reporting and audit trail.
16. QA, user acceptance testing, training, deployment and handover.

Planning note: timeline, cost, technology and milestone commitments should be estimated only after the client confirms scope and integrations.